



CLIENT DELIVERABLE

# SAOS Backup & Recovery Guide v1.0

Systack Documentation

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| ROLE                   | COLOR       | HEX                                                                                           |
|------------------------|-------------|-----------------------------------------------------------------------------------------------|
| Headers, CTAs          | Navy        |  #001a2d |
| Navy Light             | Navy Light  |  #002845 |
| Secondary accents      | Teal        |  #007da9 |
| Primary buttons, links | Cyan        |  #00a1db |
| Gradients              | Cyan Bright |  #00c5e0 |
| Backgrounds            | Gray 50     |  #f8fafc |
| Cards                  | Gray 100    |  #f1f5f9 |
| Borders                | Gray 200    |  #e2e8f0 |
| Muted text             | Gray 400    |  #94a3b8 |
| Body text              | Gray 600    |  #475569 |
| Headings               | Gray 800    |  #1e293b |
| Success                | Green       |  #22c55e |
| Error                  | Red         |  #ef4444 |

| ROLE              | COLOR  | HEX                                                                                         |
|-------------------|--------|---------------------------------------------------------------------------------------------|
| Accent highlights | Purple |  #8b5cf6 |

# SAOS Backup & Recovery Guide

**Version:** 1.0

**Date:** 2026-07-05

**Status:** Active

## Overview

SAOS uses `pg_dump`-based backups with automated restore verification to ensure data recoverability.

## Backup Strategy

| ASPECT      | CONFIGURATION                              |
|-------------|--------------------------------------------|
| Backup type | <code>pg_dump</code> (full logical export) |
| Format      | SQL (plain text)                           |
| Frequency   | Daily (manual or script)                   |
| Location    | <code>~/saos-backups/</code>               |
| Retention   | 30 days rolling                            |
| Compression | None (plain SQL)                           |
| Encryption  | Filesystem-level (AES-256)                 |

## Recovery Objectives

| METRIC                         | TARGET     | VERIFIED                                                                                                  |
|--------------------------------|------------|-----------------------------------------------------------------------------------------------------------|
| RPO (Recovery Point Objective) | 24 hours   |  24h (daily backup)  |
| RTO (Recovery Time Objective)  | 10 minutes |  ~3 minutes (tested) |

## Verification Process

The `scripts/backup_verify.py` script performs:

- pg\_dump** — Full database export
- SHA-256 checksum** — Integrity hash
- Test restore** — Create temporary database, restore backup
- Verification queries** — Check tables, record counts, data integrity
- Cleanup** — Drop test database
- Logging** — Results logged to `backup_log` table

## Running a Backup Verification

```
# Full backup + verify
python3 scripts/backup_verify.py --full

# Verify existing backup only
python3 scripts/backup_verify.py --verify-only /path/to/backup.sql
```

## Verification Checks

| CHECK               | DESCRIPTION                     |
|---------------------|---------------------------------|
| Table existence     | All 10+ expected tables present |
| Client records      | Client count matches            |
| Audit log entries   | Audit records present           |
| Compliance policies | Active policies exist           |
| RBAC roles          | Role definitions present        |

## Backup Log Table

All backup operations are logged to the `backup_log` PostgreSQL table:

| FIELD                     | DESCRIPTION                          |
|---------------------------|--------------------------------------|
| backup_type               | pg_dump, file_archive, config_backup |
| target                    | What was backed up                   |
| status                    | success, failed, verified            |
| file_path                 | Backup file location                 |
| file_size_bytes           | Backup file size                     |
| checksum_sha256           | Integrity hash                       |
| started_at / completed_at | Timing                               |
| verified_at               | When restore test ran                |
| verification_result       | Check results                        |
| rpo_minutes / rto_minutes | Recovery metrics                     |
| notes                     | Additional info                      |

## API Endpoints

| METHOD | ENDPOINT                           | AUTH       | DESCRIPTION         |
|--------|------------------------------------|------------|---------------------|
| GET    | <code>/api/admin/backup-log</code> | admin, ops | View backup history |
| POST   | <code>/api/admin/backup/run</code> | admin, ops | Trigger backup      |

| METHOD | ENDPOINT                  | AUTH       | DESCRIPTION     |
|--------|---------------------------|------------|-----------------|
| GET    | /api/admin/backup/rpo-rto | admin, ops | RPO/RTO metrics |

## Disaster Recovery

### Scenario: Database Corruption

1. Stop SAOS API service
2. Drop corrupted database
3. Restore from latest verified backup: `psql -d systack_memory -f backup.sql`
4. Restart SAOS API
5. Verify data integrity via dashboard
6. Estimated RTO: 5-10 minutes

### Scenario: Hardware Failure

1. Provision new server (or use backup hardware)
2. Install PostgreSQL, Python, dependencies
3. Restore backup from `~/saos-backups/`
4. Update Tailscale to point to new server
5. Restart SAOS API
6. Estimated RTO: 30-60 minutes (depends on provisioning)

### Scenario: Data Deletion (Accidental)

1. Identify deletion time from audit\_log
2. Find backup from before deletion
3. Restore backup to test database
4. Extract deleted records
5. Insert recovered records into production
6. Estimated RTO: 15-30 minutes

*Last updated: 2026-07-05 by SOL.*